Which Stem Carries the Era of a Song, and Can a Generator Be Steered to It?

Stem-wise probes of release decade and release market, and era-targeted accompaniment regeneration on Discogs-VI clips

James Christian* Angel Hsia†

CommE5070, Deep Learning for Music Analysis and Generation, NTU

Abstract

We ask two connected questions on the course subset (PA1) of Discogs-VI clips: which part of a recording (vocal stem, accompaniment stem, or the mix) lets frozen audio encoders recognise the release *decade* and the release *market*, and can a generator that keeps the original vocal and melody regenerate the accompaniment in a requested decade. With linear probes, stem recombination, concept erasure and signal-level controls we find that decade information is carried by the accompaniment at least as much as by the vocal, whereas market information follows the vocal stem and is largely explained by sung language. For generation we keep the original vocal, condition MusicGen-Melody on the chroma of the original mix, fine-tune a low-rank adapter on decade-labelled accompaniment and add a three-branch guidance rule that keeps the melody term and adds an era-contrast term. With codec-matched probes of two encoders, an independent third encoder, and distribution and pairing metrics, the system orders six prompted decades monotonically in expected decade and largely keeps song identity (chroma song-identification top-1 0.875 against 0.944 for neutral generations of the same adapter), while the unmodified model moves only the 1980s and 2010s. We also report where the evidence is weak: probes respond to bandwidth cues, the 1990s prompt is wording-dependent, and no listening test was run.

1 Introduction

Release decade and release market are weak, edition-level labels: in the full Discogs-VI metadata a composition has on average 2.42 distinct release countries and 1.99 distinct release decades (Sec. 2). Still, models trained on them are used as "era" or "region" detectors, and generators are asked to produce music "in the style of the 1980s". Two questions follow. (i) Which *part* of a recording, and which *cue* (sung language, genre, recording bandwidth), does a detector actually use? (ii) If a stable cue exists, can it be used to steer a cover that keeps the original melody and vocal?

The closest work we found is Lee et al. [1], who classify release era of US and Korean pop from audio and name as their main limitation that models cannot separate compositional style from recording technology. They do not separate stems. Hsieh et al. [2] introduced shuffle-and-remix of vocals with accompaniments of other songs to study confounds in singer identification; we use the same intervention for era and market. Style-transfer covers have been studied by Demerlé et al. [3], who address genre, not release era. To our knowledge, we did not find prior work that evaluates release era as a controlled attribute of a melody-preserving cover (this is the result of a literature search, not a proof of absence).

Contributions. (1) A stem-wise analysis with causal-style interventions (stem swap, linear erasure with random-rank controls, retrained-probe cue ablations) on two tasks. (2) An era-controllable cover system built from a public generator, an adapter and a guidance rule, evaluated with codec-matched probes, an independent encoder, a zero-shot audio-language judge, distribution and pairing metrics, and content-preservation metrics. (3) Controls that show what the probes also respond to (bandwidth, spectral balance, wording), reported with the main result rather than separately.

2 Data and protocol

Clips. PA1 provides 30 s mono 24 kHz excerpts from Discogs-VI recordings [4] with artist-disjoint splits. Dataset A (decade, six classes 1960s–2010s): 1026 train, 132 validation clips with labels. Dataset B (market: US, UK, Brazil, Spain, Germany, Italy; all released in the 1980s): 798 train, 102 validation. Test labels are hidden, so all accuracies are on the validation split (chance .167). The audio is YouTube-sourced and licensed for the course; we release code and result tables, not audio.

Stems. HT Demucs [5] separates vocals from drums+bass+other (accompaniment); BS-RoFormer is used as a second separator. Stems are loudness-normalised to −23 LUFS.

Probes. Frozen MERT-v2-30s [6] and MuQ [7] hid-

*UT Austin (MS in AI)

†Graduate Institute of Communication Engineering, NTU (MS)

Table 1: Validation accuracy of linear probes (A: n=132, B: n=102; chance .167).

Task	Encoder	Mix	Vocals	Accomp.	Recomb.
A decade	MERT	.477	.394	.485	.485
	MuQ	.477	.318	.424	–
B market	MERT	.588	.608	.422	.608
	MuQ	.471	.471	.392	–

Table 2: Interventions on what the probes use (MERT, validation).

Intervention	Result
Shuffle-and-remix: vocal of clip i + accompaniment of clip j, other label (A: 396 pairs, B: 306)	A: prediction = accompaniment label 34.8% [29.5, 39.9] vs. vocal label 16.4% [12.1, 21.0]. B: vocal label 50.0% [42.2, 58.2] vs. accompaniment label 18.3% [13.1, 23.5]. Same with BS-RoFormer.
LEACE erasure of the language subspace (B, rank 5)	mix .588 → .353, vocals .608 → .402, accomp. .422 → .314; random same-rank erasure: .591, .606, .413. A (85% English): no change.
English-sung clips only (B, pooled out-of-fold)	mix .397 (majority share .274) vs. .591 on all clips.
LEACE erasure of style PCs (Discogs-EffNet tags; k=10)	A mix .477 → .409 (random .49); B accomp. .422 → .294.

den states, mean-pooled over time; standardised logistic regression, layer and C chosen by cross-validation on train only, scored once on validation; 95% bootstrap intervals over clips; exact McNemar tests with Holm correction. Single-split differences below about .05 are within layer-selection noise (a 3-fold choice gives mix .523 instead of .477 on A), and intervals span about ±.09.

Label semantics. In the full Discogs-VI metadata (1.91 M versions, 348,796 cliques), 62.1% of cliques have ≥ 2 release countries and 60.0% have ≥ 2 release decades; 44.1% of 1980s versions in the six market countries sit in a clique with a 1980s version in another of the six. A release-market label is therefore a property of an edition. (PA1 clips carry no ids, so these statistics describe the full metadata, not PA1.)

3 Which stem carries the cue

Table 1 shows the pattern: for decade the accompaniment is as informative as the mix, the vocal stem is lower; for market the vocal stem is as informative as the mix and the accompaniment is lower. After Holm correction, the significant contrasts are mix vs. vocals on A with MuQ (adjusted p=.012) and, on B with MERT, mix vs. accompaniment (.038) and vocals vs. accompaniment (.022); MERT vocals vs. accompaniment on A (p=.073 before correction) is not significant. Re-summing the separated stems reproduces the mix accuracy (.485, .608), so separation itself does not change the result. BS-RoFormer reproduces the ordering (MERT, A: vocals .386, accompaniment .515; B: .549 vs. .392), but vocal-stem probes lose .08–.13 accuracy when the separator at test time differs from the one at training time, so absolute stem accuracies depend on the separator. Richer probes (MLP, SVM, layer concatenation, stem or encoder fusion), other pooling statistics, a layer-weighted sum, and five encoders (MERT, MuQ, Whisper, AST, CLAP) do not change the ordering; MERT is best or tied on every condition. Accuracy has not saturated with training size for A.

Table 2 separates cues. Swapping stems shows that the *decade* label follows the accompaniment and the *market* label follows the vocal (cluster bootstrap over vocal clips). Erasing the linear subspace of Whisper-LID language [8] groups, with a same-rank random control [9], removes most of the market accuracy while random erasure costs nothing; the result is unchanged when the language groups come from Whisper-large-v3 instead of Whisper-small (.382, .392, .314). Language labels are a proxy (no ground truth exists for these clips; the two LID models agree on 87.6% of B clips after grouping). For decade, erasing style directions lowers accuracy more than random erasure, so part of the decade signal is entangled with genre/style, but accuracy stays far above chance. Retrained-probe cue ablations indicate that decade information in the accompaniment survives band-limiting, compression, noise and segment shuffling at .36–.49 and loses most from removing content below 500 Hz (.485 → .364, p=.007), while market information in the vocal stem is unchanged by band-limiting and compression and falls only under time reversal (.608 → .480, p=.029), the pattern expected from phonetic content. Erasure bounds linear reliance from above; it does not prove that the remainder is musical [9].

4 Era-targeted cover generation

System. For each of 24 validation songs of dataset A we keep the original vocal stem and regenerate the accompaniment with MusicGen-Melody 1.5B [10] (text prompt for the era, chroma of the original mix), then remix at the original accompaniment level. A LoRA adapter [11] (r=16 on decoder attention, 3000 steps, batch 8, lr 10^{-4}, 10% neutral-prompt dropout) is trained on the accompaniment stems of the 1026 training clips with the clip's decade prompt as text. At inference we use three branches per codebook, with log-softmaxed logits

$$\ell = \ell_{\text{null}} + g_m(\ell_{\text{neut}} - \ell_{\text{null}}) + g_e(\ell_{\text{era}} - \ell_{\text{neut}}),$$

where "neut" is the neutral prompt with chroma, "era" the era prompt with chroma, g_m=3 the standard melody-guidance scale [12] and g_e the era scale; g_e=g_m

Table 3: ΔP (MERT / MuQ) for the 1960s, 1980s and 2010s prompts; song-identification top-1 (neutral LoRA generations: .944). 24 songs × 3 seeds. The LoRA row is the mean over three adapter training seeds. Intervals include 0 only for the base 1960s cells; the three-branch rows are positive in 20–24 of 24 songs ($p < .001$) at g_e=7, 10.

System	1960s	1980s	2010s	top-1
Base, std. guidance	−.001/+.006	+.593/+.414	+.328/+.094	.892
LoRA (3 train seeds)	+.392/+.255	+.253/+.207	+.120/+.077	.86–.88
+3-branch g_e=7	+.618/+.478	+.517/+.351	+.353/+.286	.875
+3-branch g_e=10	+.698/+.588	+.648/+.475	+.424/+.370	.875
+3-branch g_e=20	+.759/+.761	+.826/+.637	+.649/+.650	.840

Table 4: Six prompted decades, +3-branch g_e=10: ΔP and expected decade E (neutral E: MERT 3.05, MuQ 2.94). Base model g_e=7 in the last row.

		60s	70s	80s	90s	00s	10s
MERT	ΔP	.70	.60	.65	.26	.26	.42
	E	0.26	1.15	2.25	3.31	4.05	4.49
MuQ	ΔP	.59	.56	.48	.18	.32	.37
	E	0.65	1.43	2.38	3.08	3.91	4.30
EffNet	ΔP	.54	.28	.42	.26	.09	.18
	E	0.74	1.79	2.46	2.90	3.33	3.73
Base MERT	E	2.80	4.46	2.47	3.50	4.12	4.62

recovers standard guidance. Keeping the melody term at g_m=3 preserves song identity; contrast-only variants (negative-prompt, delta guidance) lower song-identification top-1 to .67–.68 (vs. .875).

Evaluation. Each song, prompt and generation seed gives one clip (24 songs × 3 seeds). *Era score:* decade probes (MERT layer 22, MuQ layer 9) trained on real accompaniment stems after an EnCodec [13] round trip ("codec-matched"), because probes trained on original stems barely register generated audio. The statistic is $\Delta P = P(\text{target decade} \mid \text{era prompt}) - P(\text{target decade} \mid \text{neutral prompt})$ per song, with 95% song-bootstrap intervals; E is the mean expected decade index (0 = 1960s). *Independent evaluators:* a Discogs-EffNet decade probe (own training, validation accuracy .523), zero-shot CLAP [14], and an audio-language judge [15]. *Identity:* chroma song-identification top-1 among the 24 originals (chance .042), chord agreement, tempo difference, and Discogs-VINet. *Quality and distribution:* Audiobox Aesthetics [16], a KAD-style MMD margin [17] (own implementation), FD-VGG against 171 real train clips of the prompted decade, and per-clip KL-PaSST to the original clip. COCOLA [18] measures vocal–accompaniment fit.

Main result. Table 3 shows that the unmodified model follows the 1980s prompt strongly and the 2010s prompt moderately but does not follow the 1960s prompt. The adapter makes the 1960s prompt-able; the three-branch rule adds to every decade and both encoders, and for the 1960s, 1970s, 1980s, 1990s and 2010s the gain keeps growing with g_e up to 20 (the MERT 1960s value plateaus between 15 and 20). The gain over the adapter alone replicates on two further adapter training seeds at g_e=7 and 10 (e.g. MERT 1960s: seed 1 +.658 and +.709, seed 2 +.601 and +.661). Across training seeds the adapter alone has MERT standard deviations of .024 (1960s), .015 (1980s) and .052 (2010s), so the 2010s effect is the least reproducible.

Six decades. Table 4 and Fig. 1 extend the prompts to the 1970s, 1990s and 2000s. With the adapter, E increases monotonically with the prompted decade in both encoders and in the independent EffNet probe, and every ΔP interval excludes 0 in all configurations. The base model does not: its 1970s prompt gives E=4.46 and a negative ΔP (−.036, interval excludes 0). The 1990s prompt is the weakest (MERT top-1 target share .46 at g_e=10, .58 at 15 and 20). For the 2000s prompt the primary adapter's ΔP falls from +.257 (g_e=10) to +.181 (20) while E drifts toward the 2010s (4.05 → 4.19); the two seed adapters do not show this (+.335/+.362 at 10, +.377/+.347 at 20), and a pop-punk wording of the 2000s prompt keeps its effect from g_e=10 to 15 (MERT +.240 at both), so the decline is specific to an adapter and a wording. Wording matters strongly for the 1990s: a grunge wording gives MERT ΔP=+.098 and E=3.86, a eurodance wording +.413 at g_e=15, and no wording beats the default prompt in both encoders.

Content and quality. Song-identification top-1 stays .84–.88 for the main decades at g_e=7–20 (.944 neutral), chord agreement .33–.36 (rotation floor .02). For the extra decades it falls with g_e (1970s .833 at 10, .764 at 20) and chord agreement drops to .317–.337. Audiobox production quality does not drop (3-branch: 7.50–7.52 vs. 7.31 neutral); the 2000s prompt loses 0.27 PQ only at g_e=20. KAD-style margins rise with g_e (same encoder family as the probe, so not independent). FD-VGG against real clips of the prompted decade is lower than neutral for the 1980s and 2010s in both seed adapters at g_e=7 and 10 (interval excludes 0 in 7 of 8 cells; 2010s: −1.9 to −2.1) but not for the 1960s; for $g_e \geq 15$ the 2010s gain stays significant and the 1980s gain no longer is, and 72 clips per cell keep intervals wide ($\approx \pm 1$). For the seed adapters at every guidance level and all six decades, the mean KL-PaSST to the original clip is lower than neutral (−0.18 to −0.42) and the mean CLAP text score higher (+.04 to +.26), i.e. era-prompted clips remain classified like their originals (not so for every base-model cell, e.g. 2010s KL +.025). We therefore recommend g_e=7–10, where the probes, FD, quality and identity agree (interpretation).

Other generators. MuseControlLite (Stable Audio Open with melody adapter) preserves the melody more strictly (top-1 1.000) but the same prompts barely move the probes (MERT 2010s +.171, others ≤ +.015); ACE-Step 1.5 in cover mode moves the 1960s and 1980s probes (+.05–+.08, +.13–+.22 in MERT) but keeps much less of the song (top-1 .27–.35 at every strength). Text-only MusicGen keeps .188 song-identification, showing that the chroma condition, not the kept vocal, preserves identity.

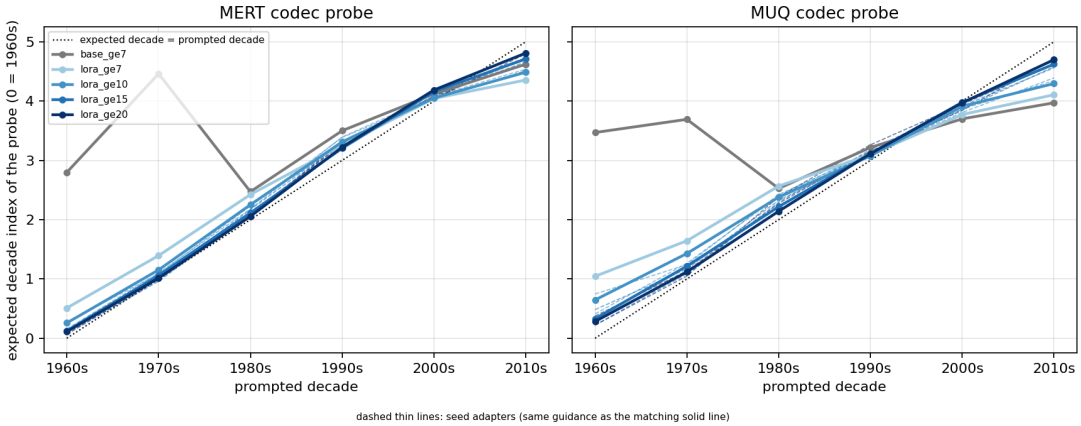


Figure 1: Expected probe decade per prompted decade (11 configurations: base, LoRA at $g_e \in \{7, 10, 15, 20\}$, and seed adapters as thin dashed lines).

5 What else moves the probes

Probe training set. Training on original stems, codec-resynthesised stems, their union, and double-codec stems gives the same sign and similar size for 1980s and 2010s (differences $\leq .09$ for double-codec); a probe trained only on original stems barely registers the generated 1980s (MERT $+.01$ to $+.23$ vs. $+.25$ to $+.83$ codec-matched), which is why we use codec-matched probes.

Bandwidth. Band-limiting the real accompaniment moves the probes toward earlier decades (MuQ $-.61$, MERT $-.30$ expected decades) and so does added hiss; soft clipping, limiting and a high shelf do not. Era prompts change high-frequency content in every system (+4 to +12 dB above 7 kHz for the 1980s prompt), with weak-to-moderate correlation between spectral change and probe change (Spearman up to .44). Matching each era clip's long-term spectrum to its neutral clip leaves MERT 1980s and 2010s effects positive (three-branch g_e=10: $+.648 \rightarrow +.553$, $+.424 \rightarrow +.406$) and the 1960s effect unchanged; the MuQ 2010s effect halves ($+.370 \rightarrow +.187$) and the base-model MuQ 2010s effect disappears ($+.094 \rightarrow -.044$, interval includes 0). Finally, prompts that name no decade but only bandwidth change the expected decade of the same generator: "dull, band-limited, tape hiss" shifts it by -1.34 (MERT) and -0.98 (MuQ) at g_e=10 (the 1960s prompt shifts it by -2.79 and -2.29), and "bright, full-bandwidth" by $+0.42$ and $+0.97$. Part of the era effect is thus carried by bandwidth; the spectrum-matching control shows it is not all of it. A hard decoder-level bandwidth constraint was not run.

Evaluator disagreement. The zero-shot audio-language judge is weak on real clips (exact-decade accuracy near chance, Spearman .45–.67 with the true decade); it confirms the 1980s effect of the base model but not the adapter's 1980s and 2010s effects. CLAP and the probes agree on the sign of 2010s shifts for guidance variants but disagree for the 1960s of some adapters. We treat the probes as proxies, not as measures of perceived era.

6 Limitations

24 validation songs, 3 seeds, one validation split per task, no artist ids (so no artist-grouped intervals; pooled out-of-fold analyses are not artist-disjoint), hidden test labels. Language and genre are proxies (Whisper LID, Discogs-EffNet tags). Probe-based era scores are proxies trained on the same real clips whose labels they predict; each conclusion is reported with at least two encoders and a third family where possible. One generator family carries the main result, a single training recipe has three adapter seeds, and several comparisons are single runs. The wording of the 1990s and 2000s prompts changes results. No listening test has been run, so we make no claim about perceived era or audio quality beyond automatic metrics.

7 Conclusion

Release decade is mostly read from the accompaniment and release market from the vocal, through sung language; both are properties of editions, and part of the decade signal travels with genre and recording bandwidth. A melody-preserving cover can be pushed along the decade axis by an adapter plus a melody-preserving era-contrast guidance rule, with probe-measured effects that replicate across adapter seeds and encoders and no measured loss of song identity at moderate guidance, but the effect partly rides on bandwidth cues and is wording-dependent for the 1990s and 2000s.

Reproducibility. Code, every result file and a worklog with each run's configuration are in https://github.com/goog-msft-fb-nflx-nvda-aapl/musicgen-steering-eval (folder `topics/era_market_cover`); `src/99_make_all_tables.sh` regenerates all tables from stored results and was run from a fresh clone

with identical output. Generated audio, embeddings and the licensed PA1 clips are not in the repository.

References

[1] Lee, Lee, Ju, Kim, Lee, and Jeong. Measuring cross-cultural style diffusion through era classification: US and Korean popular music. *arXiv preprint arXiv:2608.10980*, 2026. ISMIR 2026.

[2] Hsieh, Cheng, Fan, Yang, and Yang. Addressing the confounds of accompaniments in singer identification. *arXiv preprint arXiv:2002.06817*, 2020.

[3] Demerlé, Esling, Doras, and Genova. Combining audio control and style transfer using latent diffusion. *arXiv preprint arXiv:2408.00196*, 2024. ISMIR 2024.

[4] Recep Oguz Araz, Xavier Serra, and Dmitry Bogdanov. Discogs-VI: A musical version identification dataset based on public editorial metadata. *arXiv preprint arXiv:2410.17400*, 2024. ISMIR 2024.

[5] Simon Rouard, Francisco Massa, and Alexandre Défossez. Hybrid transformers for music source separation. *arXiv preprint arXiv:2211.08553*, 2022.

[6] Yizhi Li, Ruibin Yuan, Ge Zhang, et al. MERT: Acoustic music understanding model with large-scale self-supervised training. *arXiv preprint arXiv:2306.00107*, 2023. ICLR 2024.

[7] Haina Zhu, Yizhi Zhou, Hangting Chen, Jianwei Yu, Ziyang Ma, Rongzhi Gu, Yi Luo, Wei Tan, and Xie Chen. MuQ: Self-supervised music representation learning with mel residual vector quantization. *arXiv preprint arXiv:2501.01108*, 2025.

[8] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.

[9] Belrose et al. LEACE: Perfect linear concept erasure in closed form. *arXiv preprint arXiv:2306.03819*, 2023.

[10] Jade Copet, Felix Kreuk, Itai Gat, Tal Remez, David Kant, Gabriel Synnaeve, Yossi Adi, and Alexandre Défossez. Simple and controllable music generation. In *NeurIPS*, 2023.

[11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

[12] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[13] Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. High fidelity neural audio compression. *arXiv preprint arXiv:2210.13438*, 2022.

[14] Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Marianna Nezhurina, Taylor Berg-Kirkpatrick, and Shlomo Dubnov. Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation. *arXiv preprint arXiv:2211.06687*, 2022.

[15] Yunfei Chu, Jin Xu, Qian Yang, et al. Qwen2-audio technical report. *arXiv preprint arXiv:2407.10759*, 2024.

[16] Andros Tjandra et al. Meta audiobox aesthetics: Unified automatic quality assessment for speech, music, and sound. *arXiv preprint arXiv:2502.05139*, 2025.

[17] Yoonjin Chung, Pilsun Eu, Junwon Lee, Keunwoo Choi, Juhan Nam, and Ben Sangbae Chon. KAD: No more FAD! an effective and efficient evaluation metric for audio generation. *arXiv preprint arXiv:2502.15602*, 2025.

[18] Ruben Ciranni, Giorgio Mariani, Michele Mancusi, Emilian Postolache, Giorgio Fabbro, Emanuele Rodolà, and Luca Cosmo. COCOLA: Coherence-oriented contrastive learning of musical audio representations. *arXiv preprint arXiv:2404.16969*, 2024. ICASSP 2025.